

👉 XAI: Adversarial Attacks on Explanations

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**eXplainable
Artificial Intelligence**

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

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<https://interpretable-ml-class.github.io/>

Fooling LIME and SHAP: Adversarial Attacks on Post hoc Explanation Methods

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(Slack et al. 2020)



Motivation ---

- ML has been applied for **critical** decision making
 - ▶ Healthcare
 - ▶ Criminal justice
 - ▶ Finance
- The decision makers must clearly **understand the model behavior** to
 - ▶ Diagnose the error and potential biases
 - ▶ Decide when and how much these ML models should be trusted



Motivation ---

- Trade-off between interpretability and accuracy
 - ▶ **Simple** models can be easily **interpreted** (e.g., linear regression)
 - ▶ **Complex** but also black-box model has much **better performance** (e.g., deep neural network)
- Can a ML method be both **interpretable** and **accurate**?
- *Post hoc* explanation can seemingly solve this problem:
 - ▶ First build **complex and accurate** ML models for good performance
 - ▶ Then use post hoc explanation for model **interpretation**
- The question is:
 - ▶ How robust and reliable is the *post hoc* explanation methods?

Contribution: A Framework to 'Fool' the Post Hoc Explanation Method ---

- A novel framework that can effectively **mask the discriminatory biases** of any black box classifier
 - ▶ Fooling the **perturbation based** post hoc explanation method
 - ▶ LIME and SHAP
- Allowing an adversarial entity to control and generate an arbitrary desired explanation
- Demonstration using real-world datasets with extremely biased classifier
- Existing post hoc explanation techniques are **NOT** sufficiently robust for ascertaining discriminatory behavior of classifiers in sensitive applications

🍷 Perturbation-based Post Hoc Explanation Method ---



Preliminaries & Background ---

$$\arg \min_{g \in \mathcal{G}} L(f, g, \pi_x) + \Omega(g)$$

where the loss function L is defined as:

$$L(f, g, \pi_x) = \sum_{x' \in X'} [f(x') - g(x')]^2 \pi_x(x')$$

- f is the original classifier and x is the datapoint we want to explain
- g is the explanation we want to learn, $\Omega(g)$ is the “complexity” of g
- π is the proximity measure
- X' is a **synthetic dataset**, consisting of perturbations of x

🍷 Intuition ---



🍲 Approach: Set-up ---

Adversary would like to deploy a biased classifier f !

- **Background:** the biased model f uses sensitive attributes to make critical decisions
- **Requirement:** give access of black-box models to customers and regulators who use post-hoc explanations
- **Goal:** hide bias of the classifier f

Approach: Set-up ---

What do we need?

- Input: dataset sampled from real-world distribution
- Target Product: an adversarial classifier $\mathbf{e}$
 - ▶ $\mathbf{f}$ is the biased model to be explained, while ψ is an unbiased model

$$e(x) = \begin{cases} f(x), & \text{if } x \in \mathcal{X}_{dist} \\ \psi(x), & \text{otherwise} \end{cases}$$

Approach: OOD Detection ---

Which of the inputs belong to the real-world distribution?

- Build another classifier for OOD detection
 - ▶ Assign label “False” (not OOD) to all instances in the dataset **X**
 - ▶ Perturb all instances in **X** and assign them label “True” (OOD)
 - ★ Exceptions: instances too close to observations from **X**
 - ▶ Combine data and train OOD detection classifier

$$e(x) = \begin{cases} f(x), & \text{if } x \in \mathcal{X}_{dist} \\ \psi(x), & \text{otherwise} \end{cases}$$

Experiment: Set-up ---

90% training & 10% test

Dataset	Size	Features	Positive Class	Sensitive Feature
COMPAS	6172	criminal history, demographics, COMPAS risk score, jail and prison time	High Risk (81.4%)	African-American (51.4%)
Communities & Crime	1994	race, age, education, police demographics, marriage status, citizenship	Violent Crime Rate (50%)	White Population (continuous)
German Credit	1000	account	Good Customer	Male (69%)

🍷 Experiment: Results -- COMPAS ---



🍷 Experiment: Results -- Communities and Crime ---



🍷 Experiment: Results -- German Credit ---



Takeaway from Experiments ---

- ① Accuracy of the OOD classifier $\rightarrow$ success of the adversarial attack
- ② Adversarial classifiers to LIME are ineffective against SHAP explanations
 - Ⓐ Any sufficiently accurate OOD classifier is sufficient to fool LIME, while fooling SHAP requires more accurate classifiers
- ③ SHAP less successful when using two features $\leftarrow$ local accuracy property
 - Ⓐ Distribute attributions among several features

$$e(x) = \begin{cases} f(x), & \text{if } x \in \mathcal{X}_{dist} \\ \psi(x), & \text{otherwise} \end{cases}$$

🍷 Conclusions ---

- Main contribution: A framework for converting any black-box classifier into a *scaffolded* classifier that fools perturbation-based post-hoc explanation techniques like LIME and SHAP
- Effectiveness of this framework demonstrated on sensitive real-world data (criminal justice and credit scoring)
- Perturbation-based post-hoc explanation techniques are not sufficient to test whether classifiers discriminate based on sensitive attributes



Related Works ---

- Issues with post-hoc explanations:
 - ▶ [Doshi-Velez and Kim] identify explainability of predictions as a potentially useful feature of interpretable models.
 - ▶ [Lipton] and [Rudin] argues post-hoc explanations can be misleading and are not trustworthy for sensitive applications.
 - ▶ [Ghorbani et al.] and [Mittelstadt et al.] identified further weaknesses of post-hoc explanations.
- Adversarial explanations
 - ▶ [Dombrowski et al.] and [Heo et al.] show how to change saliency maps in arbitrary ways by imperceptibly changing inputs.

- Are the experimental results sufficient to justify the conclusions?
 - ▶ In particular, how can we explain the discrepancy in results for LIME vs. SHAP?
- What about fooling other classes of post-hoc explanation methods?
 - ▶ Past work: gradient-based methods
- Alternatively, can one design post-hoc explanations that are *adversarially robust*?

Explanations can be manipulated and geometry is to blame

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(Dombrowski et al. 2019)

Introduction ---

Motivation ---

- Understand and verify aspects of ML models - Aid decision making in high-stakes scenarios → **Reliable** explanations of models!

Can we always trust model explanations?

🍷 Summary / Contribution ---

- Manipulate explanations! - Provide a theoretical understanding of such nonrobustness and derive a bound - Introduce smoothing to increase explanation robustness!

$$\|h(p) - h(p_0)\| \leq |\lambda_{\max}| d_g(p, p_0) \leq \beta C d_g(p, p_0)$$



Background + Related Work ---

- Interpretation of Neural Networks is Fragile (Ghorbani, Abid, and Zou 2019)
 - ▶ Complex decision boundary
- The (un)reliability of saliency methods [Kindermans et al.]
 - ▶ Input invariance
- Sanity checks for saliency maps [Adebayo et al.]
 - ▶ Randomization test

- Fairwashing Explanations with Off-Manifold Detergent [Anders et al.]
 - ▶ Low-dimensional data manifold vs. High-dimensional embedding space

Methodology ---



Notation ---

- Neural network $g : \mathbb{R}^d \rightarrow \mathbb{R}^K$ with relu non-linearities
- Classifies input image x into K categories, predicted class $k = \arg \max_i g(x)_i$
- Explanation map: $h : \mathbb{R}^d \rightarrow \mathbb{R}^d$

- Target map: $h^t \in \mathbb{R}^d$
- Manipulated image: $x_{adv} = x + \delta x$

🍷 Properties of Manipulated Image ---

- ① The output of the network stays approximately constant, i.e. $g(x_{adv}) \approx g(x)$.
- ② The explanation is close to the target map, i.e. $h(x_{adv}) \approx h^t$.
- ③ The norm of the perturbation δx added to the input image is small, i.e. $\|\delta x\| = \|x_{adv} - x\| \ll 1$ and therefore not perceptible.

Explanation Methods ---

Propagation-based Gradient-based

- **Vanilla gradients:** $h(x) = \frac{\partial g}{\partial x}(x)$ - Quantifies how infinitesimal perturbations in each pixel change the prediction - **Gradient $\times$ Input:** $h(x) = x \odot \frac{\partial g}{\partial x}(x)$ - For linear models, this measure gives the exact contribution of each pixel to the prediction - **Integrated Gradients:** $h(x) = (x - \bar{x}) \odot \int_0^1 \frac{\partial g(\bar{x} + t(x - \bar{x}))}{\partial x} dt$
- **Guided Backpropagation** - **Layer-wise Relevance Propagation** - **Pattern Attribution** - Standard backpropagation upon element-wise multiplication of the weights with learned patterns

Manipulation Method ---

Obtain manipulated images by optimizing the loss function:

$$\mathcal{L} = \|h(x_{adv}) - h^t\|^2 + \gamma \|g(x_{adv}) - g(x)\|^2$$

with respect to x_{adv} using gradient descent

- $h(x_{adv})$: manipulated explanation map
- h^t : target map
- γ : weighting hyperparameter
- $g(x_{adv})$: network output (manipulated input)
- $g(x)$: network output (original input)

🍷 Manipulation Method: Softplus Trick ---

- The gradient w.r.t. the input $\nabla h(x)$ of the explanation often depends on the vanishing second derivative of the relu non-linearities. This causes problems during optimization:

$$\partial_{x_{adv}} \|h(x_{adv}) - h^t\|^2 \propto \frac{\partial h}{\partial x_{adv}} = \frac{\partial^2 g}{\partial x_{adv}^2} \propto \text{relu}'' = 0$$

- **Solution:** replace relu with softplus

$$\text{softplus}_{\beta}(x) = \frac{1}{\beta} \log(1 + e^{\beta x})$$

Experiments ---

Experimental Setup ---

- Apply algorithm to 100 randomly selected images for each explanation method
- Use VGG-16 network pre-trained on ImageNet
- For each run, randomly select two images from the test set:
 - ▶ One of the two images is used to generate a target explanation map
 - ▶ The other image is perturbed by the algorithm with the goal of replicating the target using a few thousand iterations of gradient descent
- Comparable results obtained for ResNet-18, AlexNet, and Densenet-121 + CIFAR-10 dataset

🍷 Results: Visual Comparison ---



🍷 Results: Quantitative Metrics ---

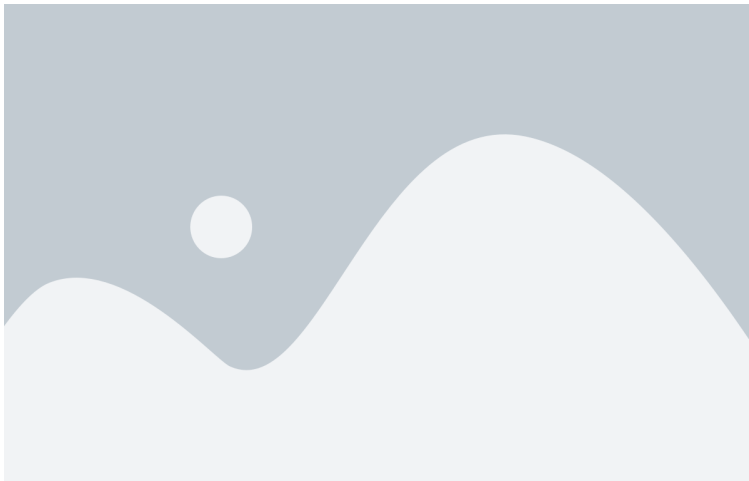


Theoretical Analysis ---

🍷 Intuition ---

Why are explanations vulnerable and unreliable?

Large curvature of the NN output manifold!



Theorem 1

Let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be a network with softplus_β non-linearities and $\mathcal{U}_\epsilon(p) = \{x \in \mathbb{R}^d; \|x - p\| < \epsilon\}$ an environment of a point $p \in S$ such that $\mathcal{U}_\epsilon(p) \cap S$ is fully connected. Let g have bounded derivatives $\|\nabla g(x)\| \geq c$ for all $x \in \mathcal{U}_\epsilon(p) \cap S$. It then follows for all $p_0 \in \mathcal{U}_\epsilon(p) \cap S$ that

$$\|h(p) - h(p_0)\| \leq |\lambda_{\max}| d_g(p, p_0) \leq \beta C d_g(p, p_0)$$

where $\lambda_{\max}$ is the principle curvature with the largest absolute value for any point in $\mathcal{U}_\epsilon(p) \cap S$ and the constant $C > 0$ depends on the weights of the neural network.

🍷 Robustness via Smoothing ---



$$\text{softplus}_{\beta}(x) = \frac{1}{\beta} \log(1 + e^{\beta x})$$

Replacing relu with softplus reduces the curvature $\rightarrow$ more robust explanations

Smoothing: Connections to SmoothGrad ---

Theorem 2

For a one-layer neural network $g(x) = \text{relu}(w^T x)$ and its β -smoothed counterpart $g_\beta(x) = \text{softplus}_\beta(w^T x)$, it holds that

$$\mathbb{E}_{\epsilon \sim p_\beta} [\nabla g(x - \epsilon)] = \nabla g_{\frac{\beta}{\|w\|}}(x)$$

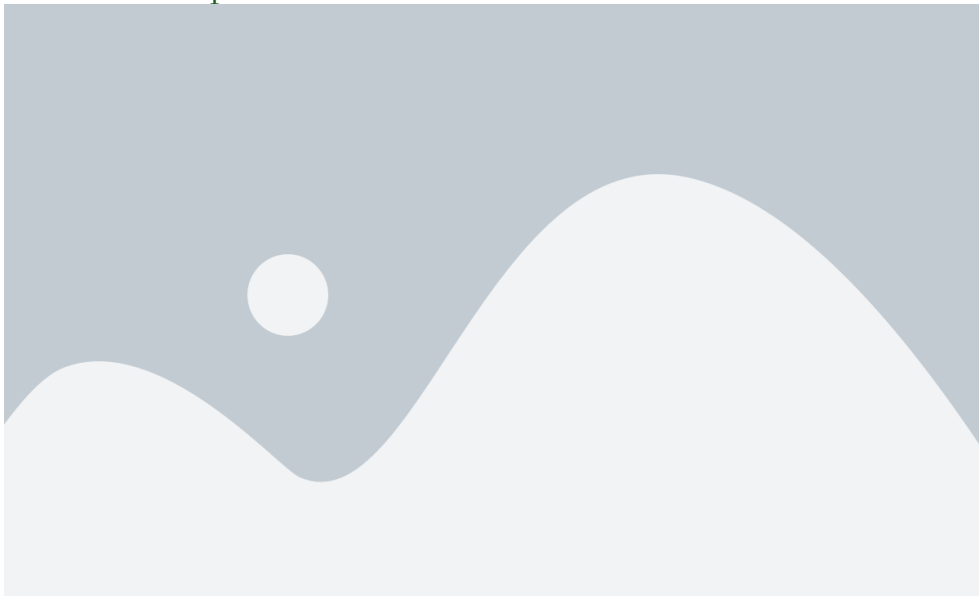
where $p_\beta(\epsilon) = \frac{\beta}{(e^{\beta\epsilon/2} + e^{-\beta\epsilon/2})^2}$.

- **SmoothGrad** $\equiv$ β -smoothing
- $\epsilon_i \approx \mathcal{N}(0, \sigma)$ with variance $\sigma = \log(2) \frac{\sqrt{2\pi}}{\beta}$

🍷 Robustness Experiments ---



🍷 Robustness Experiments ---



Conclusion ---

Critique ---

Strengths

- Thorough investigation: problem $\rightarrow$ reason $\rightarrow$ solution
- Extensive validation: various explanation methods, models, and datasets

Limitations

- Analyses focused on relu/softplus activation function
- Evaluation of robustness based on Pearson correlation coefficient

🍷 Future Directions & Food for Thought ---

Future directions

- Extend empirical analyses to other tasks and data modalities
- Generalize theoretical analyses to propagation-based methods
- Modify model training process to make NNs less vulnerable to explanation manipulation
 - ▶ Low-curvature models [Srinivas et al., NeurIPS 2022]

Food for thought

- Might there be other reasons for explanations being sensitive to manipulation?
- What are other ways to evaluate robustness of explanations?
- Is it a good idea to trade-off faithfulness for better robustness?

Thank You!



References --- |

- Dombrowski, Ann-Kathrin, Maximilian Alber, Christopher J. Anders, Marcel Ackermann, Klaus-Robert Müller, and Pan Kessel. 2019. “Explanations Can Be Manipulated and Geometry Is to Blame.” In *Advances in Neural Information Processing Systems*.
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